

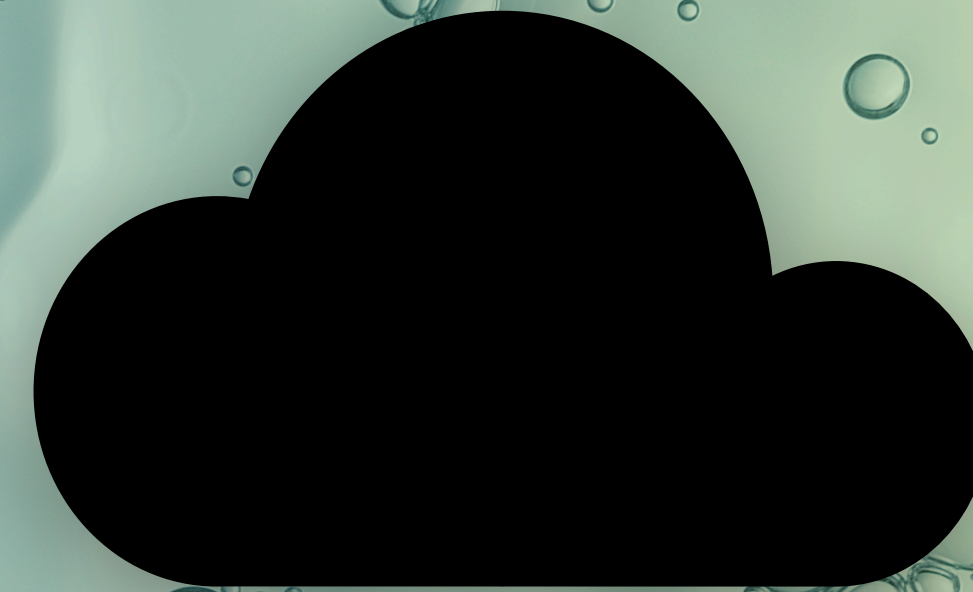
# Semester Project Weather App







- ✱ Uses OpenWeather API key as an environmental variable
- ✱ Takes text, finds the best location-based result, converts into coordinates
- ✱ Fetches:
  - ✱ Current conditions
  - ✱ 3-hour-block hourly forecast
  - ✱ 5 day forecast



- ✱ Fastest way to interact with the app
- ✱ Insert a location (--city) or be prompted for a location
- ✱ Change measurement (--units)
- ✱ Prints all data or any errors that may have occurred



- ✱ Most readable way to interact with the app
- ✱ Insert a location
- ✱ Change between imperial and metric system (F/C)
- ✱ View current conditions
  - ✱ Background dynamically changes with current conditions.
- ✱ Switch between viewing hourly and daily data



I. User location input is read and sent to OpenWeather to determine a list of possible locations

II. Conditions/Forecasts are received from OpenWeather based on the first possible location in the list

III. The location and conditions/forecasts are stored in immutable dataclasses

```
@dataclass(frozen=True)
class Location:
    name: str
    country: str | None
    state: str | None
    lat: float
    lon: float

    @property
    def display_name(self) → str:
        parts: list[str] = [self.name]
        if self.state:
            parts.append(self.state)
        if self.country:
            parts.append(self.country)
        return ", ".join(parts)
```

```
@dataclass(frozen=True)
class CurrentConditions:
    location_name: str
    temperature: float
    feels_like: float | None
    humidity: int | None
    wind_speed: float | None
    status: str
    icon: str | None
```